# BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
# PRACTICAL JOURNAL

**Class: TYIT** **Sem: VI** **Roll No.: TYIT25** **Date: 10-01-2026**

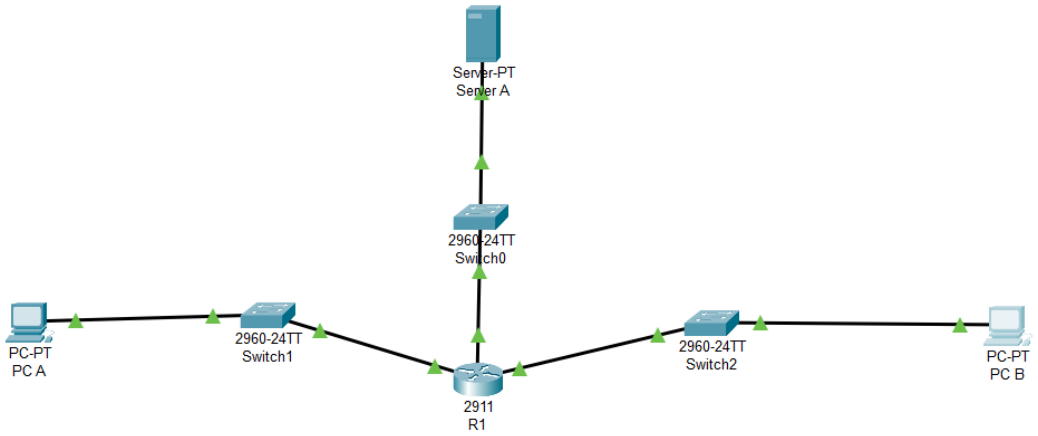
**Course Name: Cybersphere & Security (CSS)** **Page no:**

**Practical Number:- 05**

## Q). Configuring Extended ACLs.

**Steps:**

1. Create a Network Topology:
   - Open Cisco Packet Tracer and place the following devices:
     1. 1 x Server (Server-PT)
     2. 3 x Switches (2960-24TT)
     3. 2 x PC
     4. 1 x Router (1941)
   - Rename the devices and connect them using Copper straight-through wire.




2. Before applying security rules, you must configure the IP addresses for all devices to ensure connectivity.

**1. Router (R1) Interface Configuration**

Access the **Config** or **CLI** tab on **R1** and set up the three gateway interfaces:

- **GigabitEthernet0/0:** Set IP to `172.22.34.65`, Subnet Mask `255.255.255.224`.
- **GigabitEthernet0/1:** Set IP to `172.22.34.97`, Subnet Mask `255.255.255.240`.
- **GigabitEthernet0/2:** Set IP to `172.22.34.1`, Subnet Mask `255.255.255.192`.
- Ensure all interfaces are turned **On** (Status: Up).

**Teacher's Signature**

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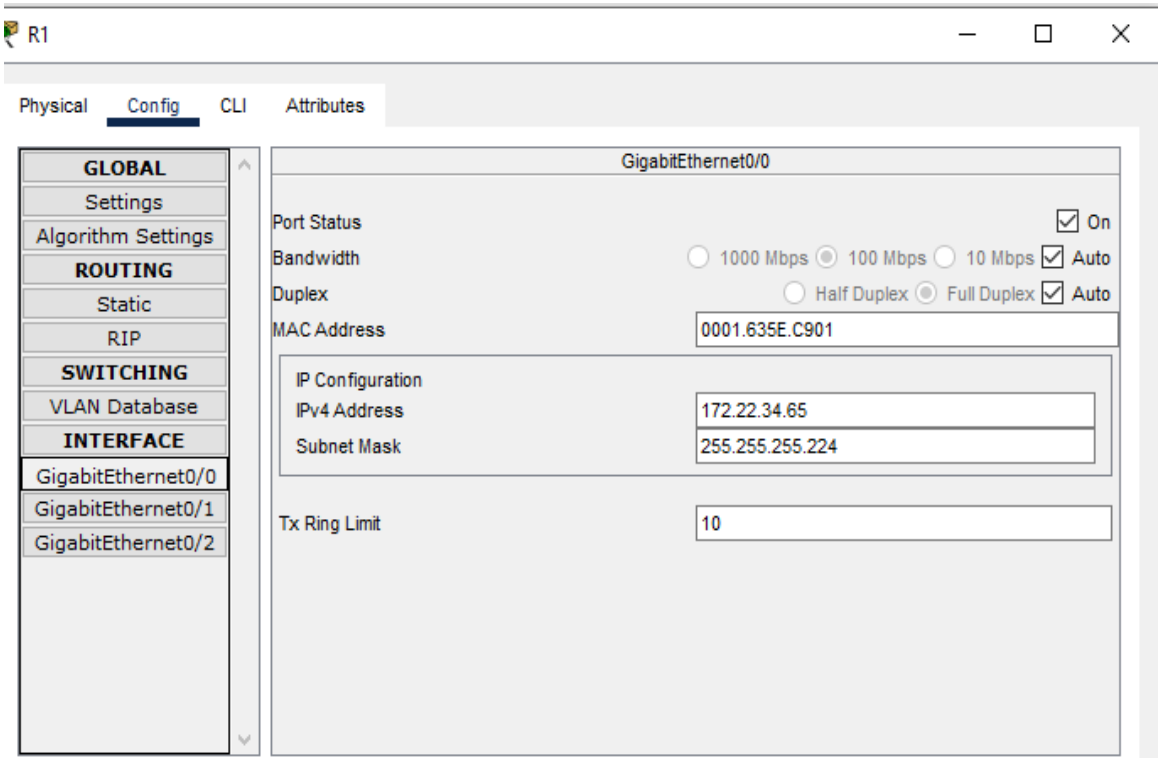
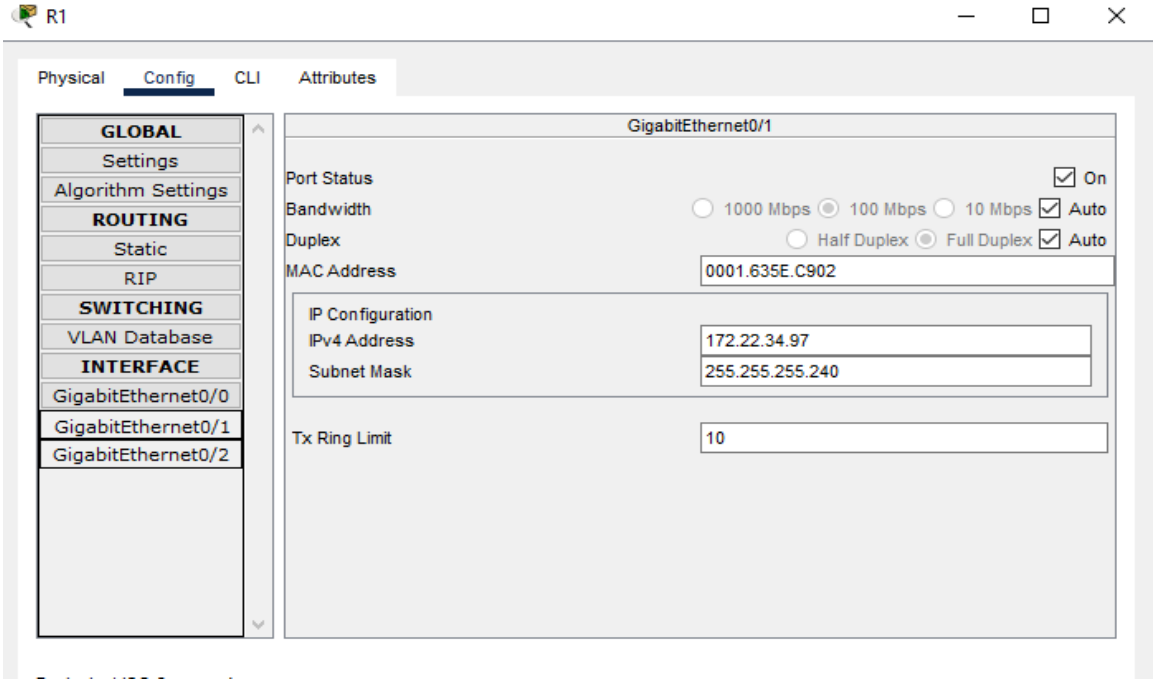
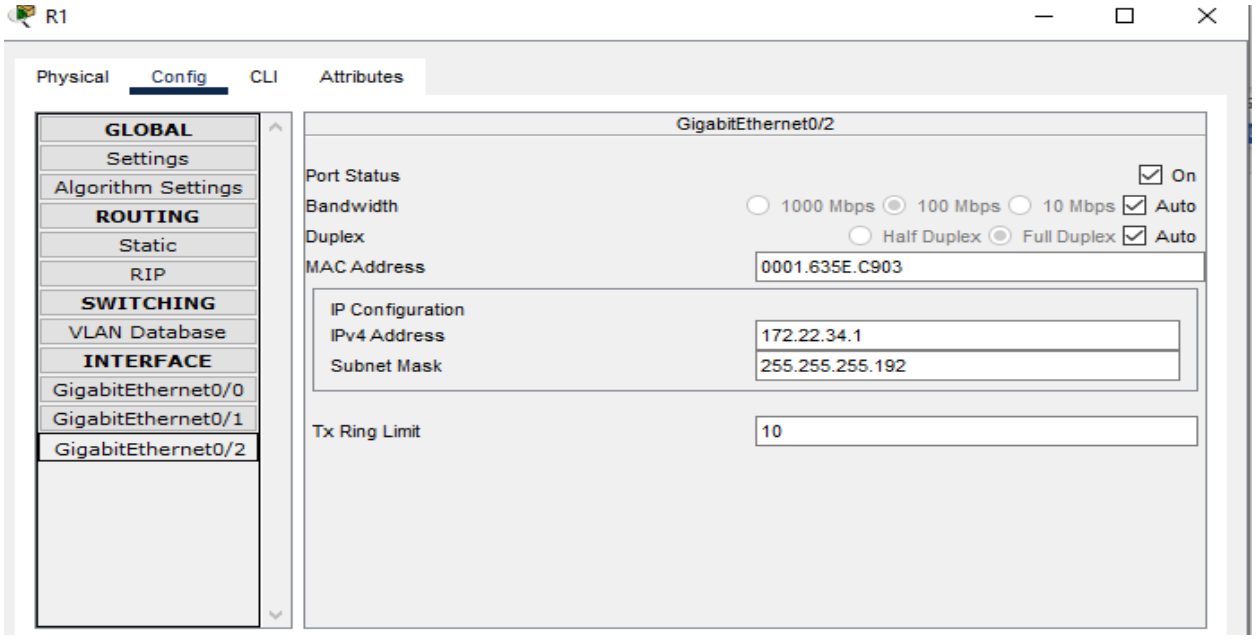


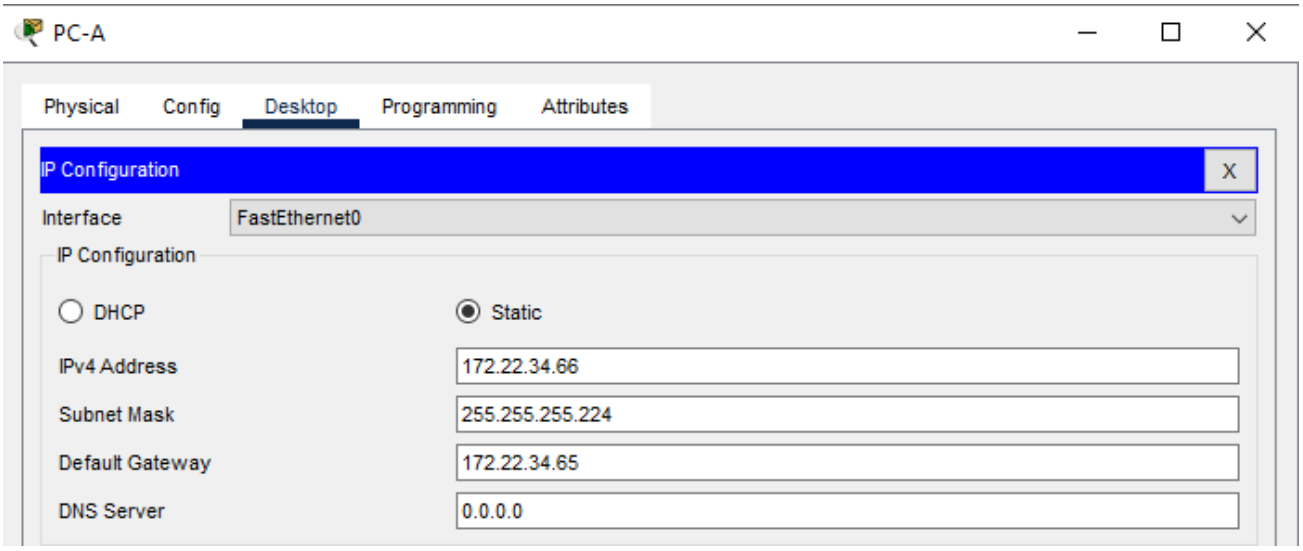

**Teacher's Signature**

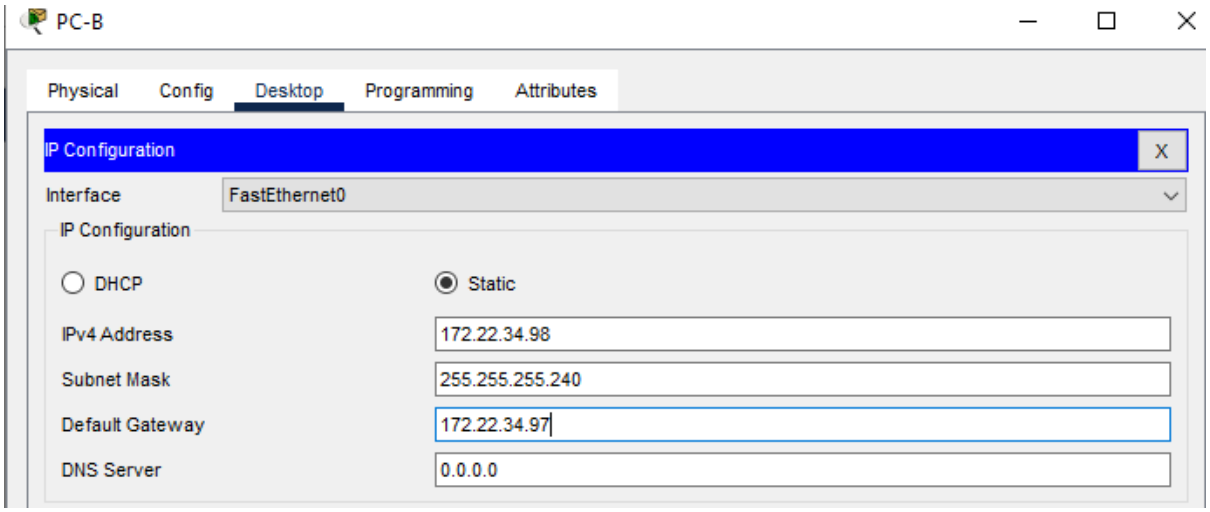


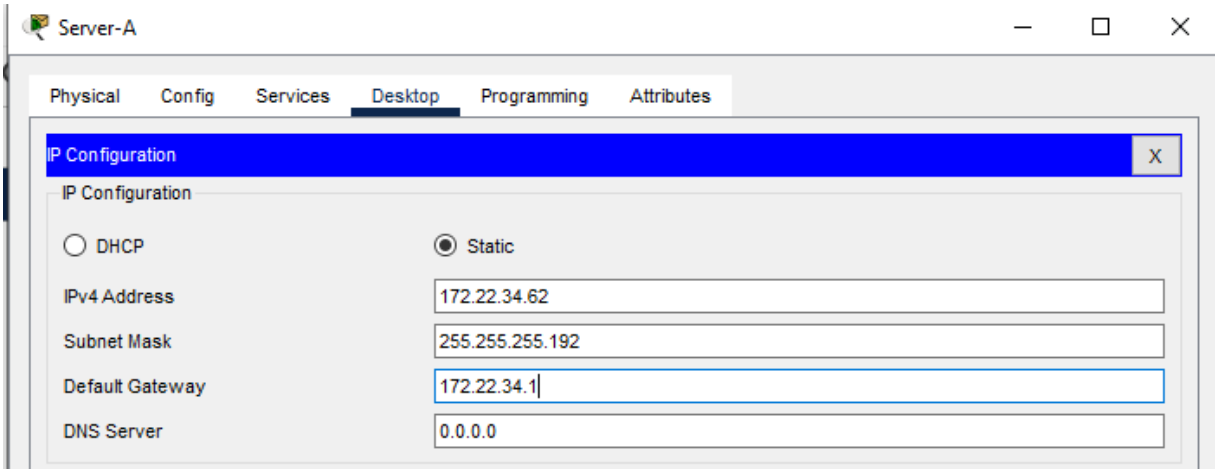

**2. End Device Configuration**

Navigate to the **Desktop > IP Configuration** tab for each device:

- **PC-A:** IP `172.22.34.66`, Subnet `255.255.255.224`, Gateway `172.22.34.65`.
- **PC-B:** IP `172.22.34.98`, Subnet `255.255.255.240`, Gateway `172.22.34.97`.
- **Server-A:** IP `172.22.34.62`, Subnet `255.255.255.192`, Gateway `172.22.34.1`.






3. **Configuring Extended ACLs**.

Extended ACLs are used here to permit specific traffic (FTP, HTTP, ICMP) and deny everything else by default.

A. Numbered ACL for PC-A (FTP & ICMP):

This ACL allows PC-A's network to access Server-A via FTP and Ping.

- **Command:** `access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp`
- **Command:** `access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62`

- **Apply to Interface:** Go to interface **GigabitEthernet 0/0** and enter `ip access-group 100 in`.

```
R1(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp
R1(config)#access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip access-group 100 in
R1(config-if)#
```

B. Named ACL for PC-B (HTTP & ICMP)

This ACL allows PC-B's network to access Server-A via Web (HTTP) and Ping.

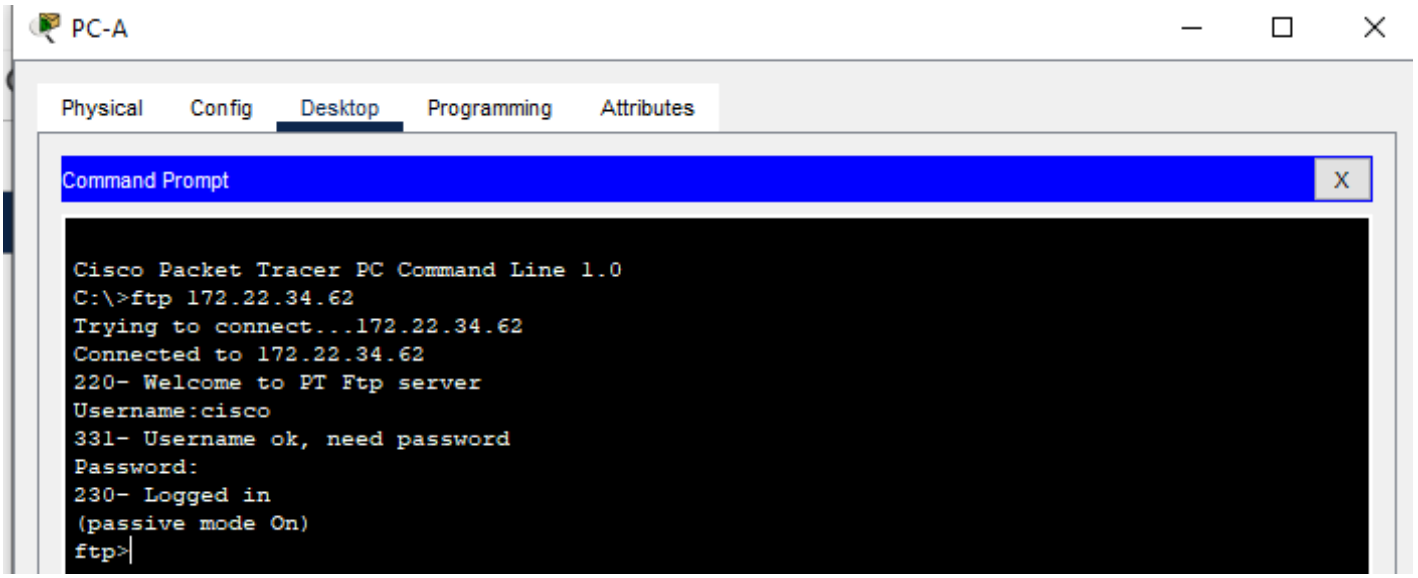
- **Command:** `ip access-list extended HTTP_ONLY`
- **Permit HTTP:** `permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www`
- **Permit Ping:** `permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62`
- **Apply to Interface:** Go to interface **GigabitEthernet 0/1** and enter `ip access-group HTTP_ONLY in`.

```
R1(config-if)#en
% Ambiguous command: "en"
R1(config)#ip access-list extended HTTP_ONLY
R1(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www
R1(config-ext-nacl)#permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62
R1(config-ext-nacl)#ex
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#ip access-group HTTP_ONLY in
R1(config-if)#
```

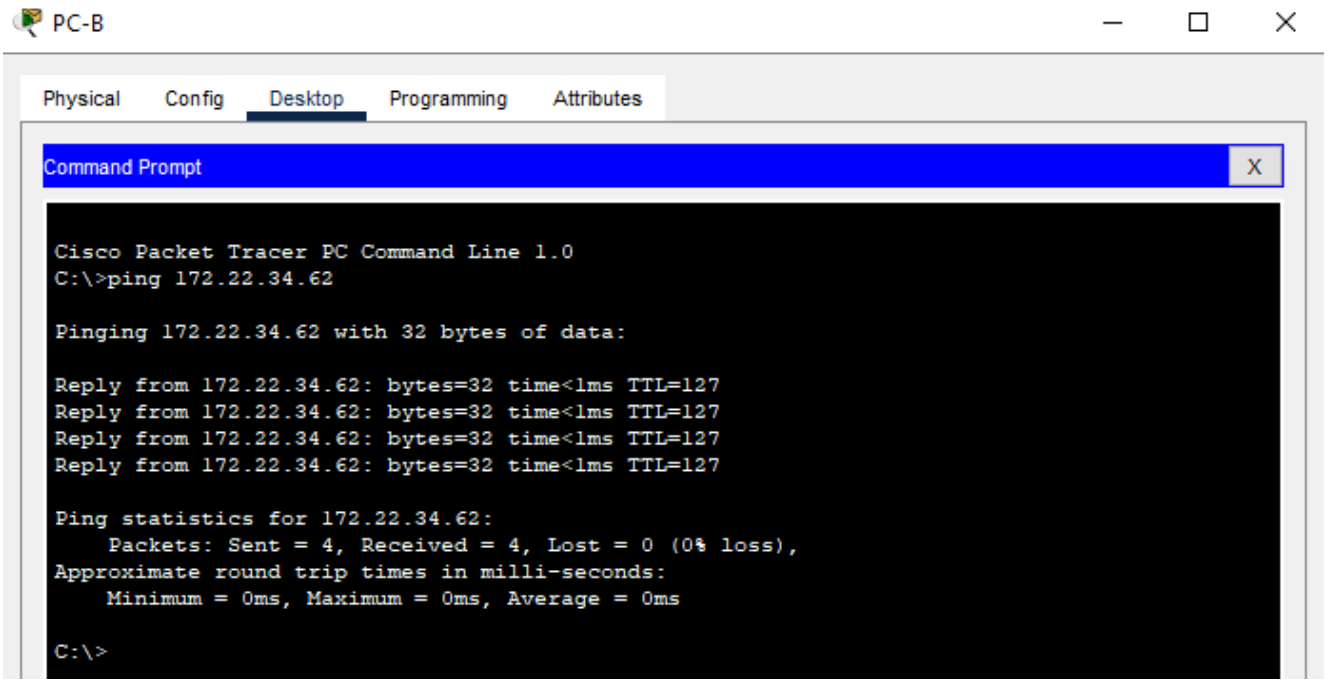
3. Once the ACLs are applied, verify that the rules are working as intended.

1. **Verify FTP Access from PC-A**

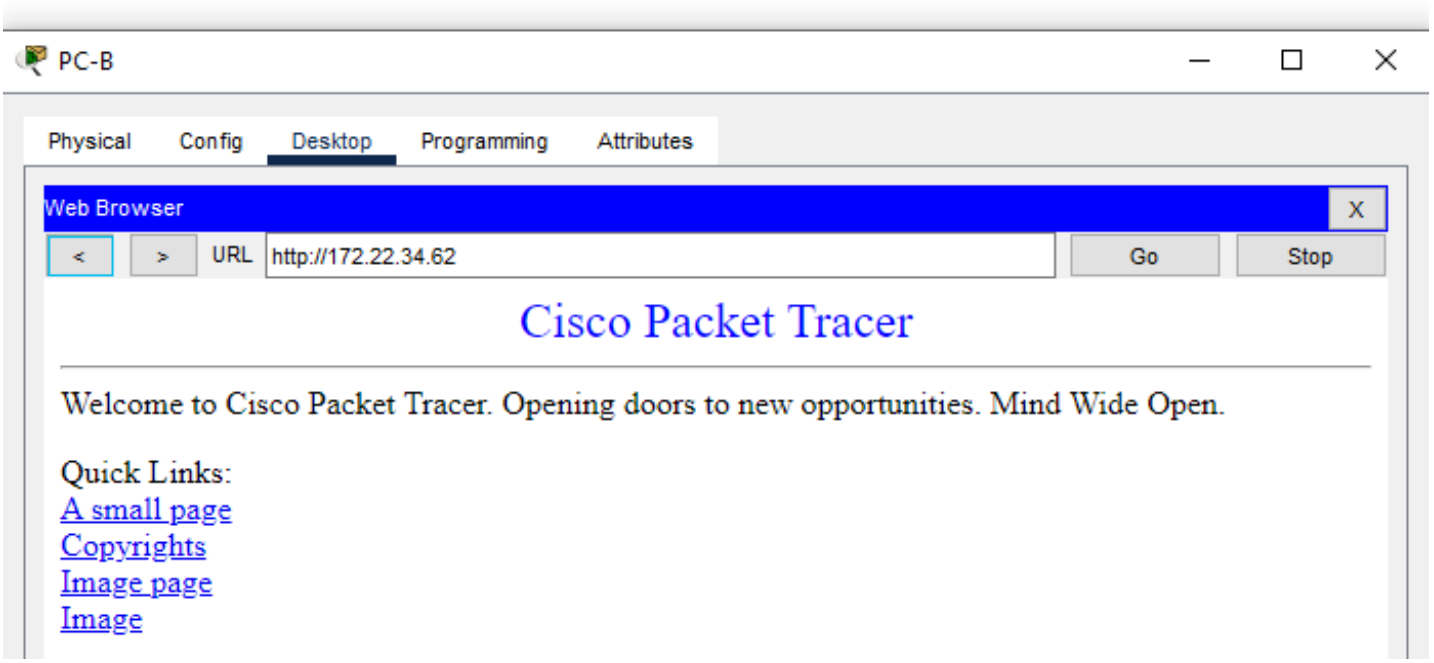
- Open **PC-A > Desktop > Command Prompt**.
- Type `ftp 172.22.34.62`.
- The screen should show "Connected to 172.22.34.62" and prompt for a username.



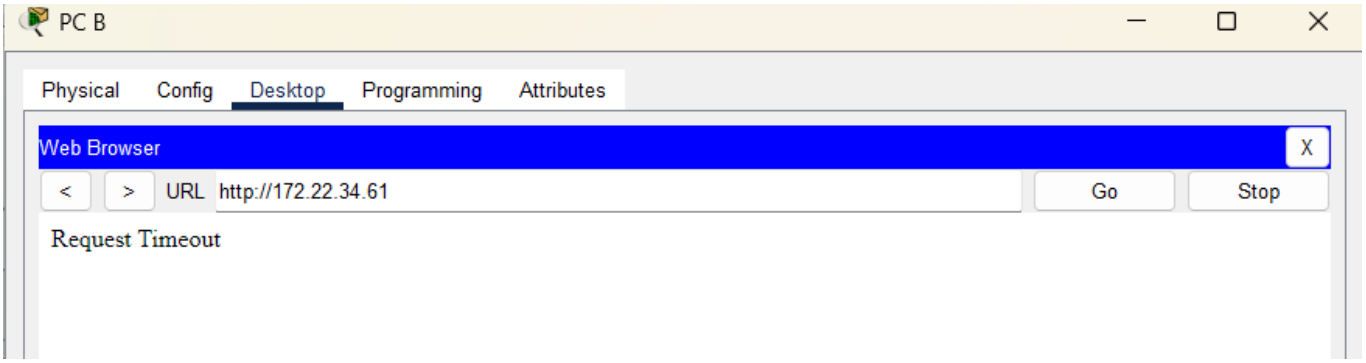

2. **Verify ICMP (Ping) Access from PC-B**
   - Open **PC-B > Desktop > Command Prompt**.
   - Type `ping 172.22.34.62`.
   - You should receive successful replies from the server.



3. **Verify HTTP Access from PC-B**
   - Open **PC-B > Desktop > Web Browser**.
   - Enter `http://172.22.34.62` in the URL bar.
   - The "Cisco Packet Tracer" welcome page should load successfully.




   - Check what will happen when a wrong URL is entered(e.g., http://172.22.34.61).




**Teacher's Signature**